Rocket Science and Technology 4363 Motor Ave., Culver City, CA 90232
Phone: (310) 839-8956 Fax: (310) 839-8855

Static Aeroelastic Effects on Fins with Spin Wedges

05July 2013

by C. P. Hoult

Introduction

The design of the second stage fins of the Nike Cajun and Nike Apache sounding rockets was extremely simple. The key component was a Al extrusion that had a single fin panel in its center integrally attached to one quarter of the fin can. After extrusion each piece had its fin panel portion cut back to generate a swept leading edge, and piano hinge fingers cut into the stream wise edges if the fin can. Further assembly included attachment of an Inconel fin leading cuff for thermal reasons, and attaching the four individual fin panels with wire pins through the piano hinges. It was a simple and inexpensive design.

There was only one problem: there was no direct way to adjust the second stage roll rate to manage the pitch roll resonance process. Eventually, the industry adopted spin wedges. After measuring the as-built fin cant angles a custom wedge was cut for each fin panel to zero out pitch and yaw misalignment torques, and to give the desired roll rate. This had only one problem: it did not provide an experimental roll rate history that could be connected with the theoretical history. This is fairly well documented in ref's. (1) and (2). These also supply some spurious discussion about the influence of a thick boundary on a spin wedge.

This note provides an analysis how a spin wedge can act to twist a fin plate. It is a rough approximation only intended to provide an indication of trends.

Nomeclature

Mnemonic	Definition
b	Fin semi span, ft from body centerline
$C_{N\alpha}$	Fin average airfoil normal force coefficient slope, rad^{-1}
C_P	Pressure coefficient $=\Delta p / q$
c	Local fin chord, ft
c_R	Root chord, ft
c_T	Tip chord, ft
G	Torsional modulus of elasticity of the fin material, lb/ft^2
J	Torsional stiffness of a fin airfoil, ft^4
M	Free stream Mach number
q	Dynamic pressure, lb/ft^2
Δp	Local pressure – free stream pressure, lb/ft^2
R	Body radius, ft
T	Torque about the elastic axis, ft-lb

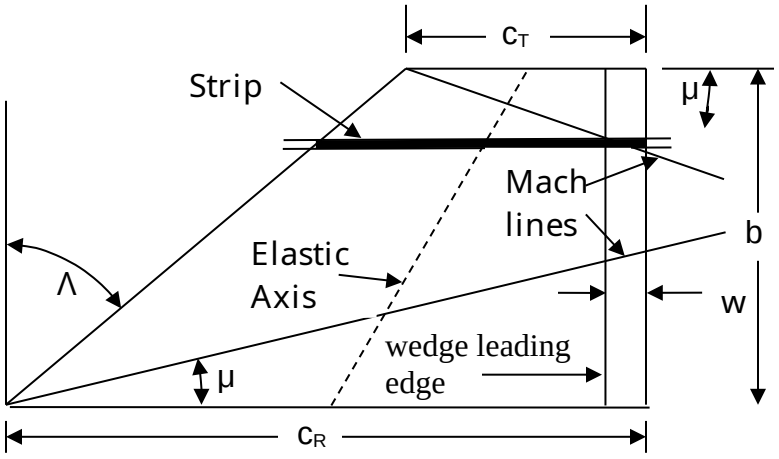
Mnemonic	Definition
t	Fin thickness, ft
U	Flight speed, ft/sec
w	Wedge chord, ft
y	Distance from center line to a fin strip, ft
α	Local angle of attack, rad
δ	Wedge ramp angle, rad
μ	Mach angle, rad
Λ	Leading edge sweep angle, rad
θ	Torsional deflection (twist angle), rad
ω	Roll rate, rad/sec

Analysis

First, the assumptions. The fin planform is assumed to be a clipped delta as sketched. Next, the Mach number is well into the supersonic regime. All angles of attack are small compared to unity. The fin airfoil shape is a flat plate with a radiused leading edge. As is customary, the roll moment of inertia is negligibly small.

The simplified aerodynamic model used here is strip theory. In this approach the local normal force coefficient slope is approximated by an average taken over the entire planform. Considering the very approximate nature of the elastic twisting model employed, it does not seem worthwhile to expend much effort on refining the airload distribution. The spin wedges are assumed to have full span wise coverage as sketched below.

The elastic model used considers the fin to be a flat plate of high aspect ratio.



Wedge Aerodynamics

Each wedge has a constant chord with a customized ramp angle designed to null out pitch and yaw torques due to fin misalignment while providing a roll driving torque needed to achieve a planned roll rate. Since wedge angles of the order of 15° are not uncommon the third order result of ref. (4) is needed to achieve a modest degree of fidelity. The wedge center of pressure is assumed to be at the wedge half chord point. Neither wedge chord nor wedge ramp angle vary with fin span wise coordinate.

Torsional Deflections

Consider the fin to be a one-dimensional beam subject to torsional deformations. Assume the elastic axis lies on the fin half chord line with no change due to the presence of a wedge. Since supersonic flow has been assumed, the aerodynamic loading due to cant angle and roll damping are assumed to have no twisting torque because the airfoil center of pressure also lies on the half chord line. Then, the main torque on a strip of fin about the elastic axis will be that due to the spin wedge:

$$dT = qC_p w(1/2\,c - 1/2\,w)dy = 1/2\,qC_p w(c - w)dy$$

The change in twist angle over the strip is

$$dT = GJ\, d/dy(d\theta/dy)\,dy = GJ\,\frac{d\theta'}{dy}dy$$

where $J = 1/3ct^3$

Putting these pieces together gives

$$d\theta' = \frac{3qC_p w}{2Gt^3}\left[\frac{c - w}{c}\right]dy \qquad (1)$$

The term in square brackets at the end of eq. (1) is only a little less than unity. It can be approximated by a constant evaluated at its half span point:

$$\varphi = \left[\frac{c - w}{c}\right]_{1/2}.$$

Then,

$$\frac{d\theta'}{dy} = \frac{3qC_p w\varphi}{2Gt^3} \qquad (2)$$

The first thing to be noticed about eq. (2) is for a fin of constant thickness the twist angle is approximately a quadratic function of the span wise coordinate. That is,

$$\theta' = \frac{3qC_p w\varphi}{2Gt^3}(y - R), \text{ and}$$

$$\theta = \frac{3qC_p w\varphi}{4Gt^3}(y - R)^2 \qquad (3)$$

Roll Equilibrium

Roll equilibrium is achieved when the three roll torques are in balance. Roll damping torque can be estimated from strip theory. A strip of span dy has

$$\text{d(Roll damping moment)} = qcdyyC_{N\alpha} * \frac{-\omega y}{U} = -qcy^2 C_{N\alpha}\frac{\omega}{U}dy$$

The chord varies according to $c = \frac{c_R b - c_T R}{b - R} - \frac{c_R - c_T}{b - R}y = A - By$ (4)

Note that ref. (5) provides the supersonic aerodynamics of fins.

Roll damping moment

$$= -qC_{N\alpha}\frac{\omega}{U}\int_R^b (A - By)y^2 dy = -qC_{N\alpha}\frac{\omega}{U}\left[A(b^3 - R^3)/3 - B(b^4 - R^4)/4\right] \quad (5)$$

$\text{d(Roll driving moment due to twisting)} = qcdyyC_{N\alpha}\theta(y)$, or

$$\text{Roll driving moment due to twisting} = \frac{3q^2 C_P C_{N\alpha} w\varphi}{4Gt^3}\int_R^b (A - By)(y - R)^2 y dy, \text{ or}$$

Roll driving moment due to twisting

$$= \frac{3q^2 C_P C_{N\alpha} w\varphi}{4Gt^3}\left[AR^2(b^2 - R^2)/2 - (2AR + BR^2)(b^3 - R^3)/3 + (A + 2BR)(b^4 - R^4)/4 - B(y^5 - R^5)/5\right]$$

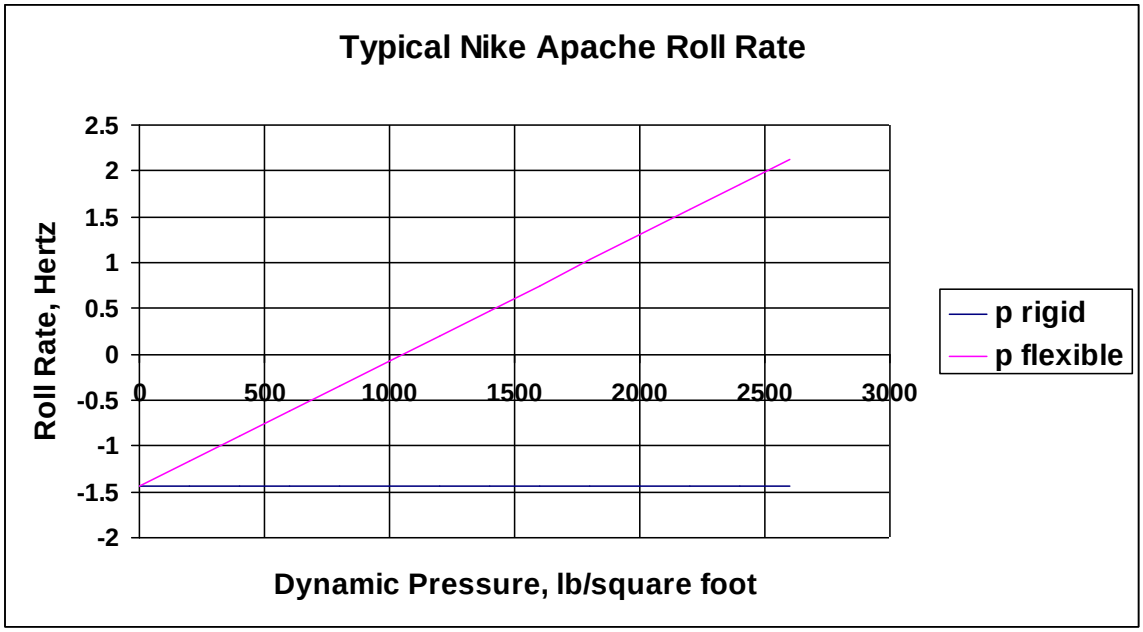
$$\text{d(Roll driving moment due to wedge)} = -qC_P wdyy$$

$$\text{Roll driving moment due to wedge} = -qC_P w(b^2 - R^2)/2$$

The equilibrium spin rate is estimated by setting sum of all the roll moments equal to zero, and solving for the roll rate.

Results

The model described above has been coded into Excel™ and filed as FIN WEDGE.xls Now, using representative data from ref. (2), The plot below can be built for a representative Nike-Apache with constant thickness fins in the middle of its second stage burn. The actual flight dynamic pressure is in the neighborhood of about 2600 psf. We first notice that the flexible fin roll rate sign reverses at about 1000 psf...so called aileron reversal. Since very few (if any) of these rockets was instrumented with a roll rate gyro, all the reported roll rates were derived from telemetry antenna pattern data which cannot measure the sign of the roll rate. In other words, the problem was more severe than believed at the time. By the way, the large torsional deflections probably also exacerbated any fin pitch or yaw misalignment torques present.



References

1. R. J. Hawks, "A Theoretical Investigation of the Rolling Motion of the Nike Cajun and Nike Apache Rocket Vehicles, NASA TM X-55282, August 1965.
2. H. L. Galloway, "The Effect of a Fin-Trailing Edge Wedge on the Roll History of a Nike Apache", NASA TM X-55463, February 1966
3. C. P. Hoult, "Normal Force and Center of Pressure of a Clipped Delta Fin with Supersonic Leading Edges", RST memo, 2013
4. Anon., "Equations, Tables and Charts for Compressible Flow", NACA Report 1135, 1953
5. C. P. Hoult, "Normal Force and Center of Pressure of a Clipped Delta Fin with Supersonic Leading Edges", RST memo, 2013